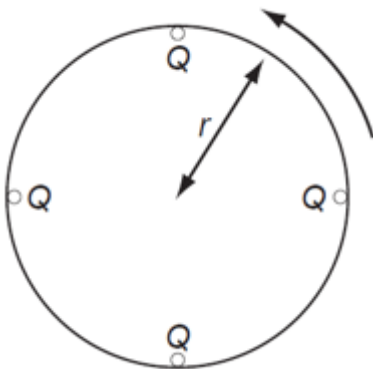


23. Four point charges, each of charge Q , are placed on the edge of an insulating disc of radius r .

The frequency of rotation of the disc is f .



What is the equivalent electric current at the edge of the disc?

- A** $4Qf$ **B** $\frac{4Q}{f}$ **C** $8\pi rQf$ **D** $\frac{2Qf}{\pi r}$

Answer: A

Consider a fixed point, $4Q$ will pass through the point in one period.

$$\text{So, current} = \frac{\text{total charge}}{\text{time}} = \frac{4Q}{1/f} = 4Qf$$